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B.Tech.-Third Semester(Backlog)
End Semester Examination, November-December, 2024

Course Code: ICMTCC04/EEMTC04
 Time: 03 hours.

Course Title: MATHEMATICS III
 Maximum Marks. 50

Note: Missing data/information(if any), may be suitably assumed and mentioned in the answer.

Q. No.	Question	Marks	CO
Q1	Attempt any 2 parts of the followings:		
1a	(i) Evaluate $\int_C \frac{\cos(\pi z^2)}{(z-1)(z-2)} dz$, where $C = \{z \in \mathbb{C} : z = \frac{3}{2}\}$. (ii) Discuss the nature of the singularity of the function $\frac{e^z}{z+4}$.	3+2=5	CO1
1b	Find the values of a and b such that the function $f(z) = x^2 + ay^2 - 2xy + i(bx^2 - y^2 + 2xy)$ is analytic. Also find $f'(z)$.	5	CO1
1c	Evaluate $\int_C \frac{z-1}{(z+1)^2(z+2)} dz$, where C is the circle $ z-i =2$, using Cauchy's residue theorem.	5	CO1
Q2	Attempt any 2 parts of the followings:		
2a	Use Stoke's theorem to evaluate $\int_C [(x+y)dx + (2x-z)dy + (y+z)dz]$, where C is the boundary of the triangle with the vertices $(2, 0, 0)$, $(0, 3, 0)$ and $(0, 0, 6)$.	5	CO2
2b	(i) Find the unit normal vector to the surface of $x^3 + y^3 + 3xyz$ at the point $(1, 2, -1)$. (ii) Show that the vector $\vec{F} = (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k}$ is irrotational.	2.5+2.5=5	CO2
2c	By Green's theorem, evaluate $\int_C e^{-x}(\sin y dx + \cos y dy)$, where C being the rectangle with vertices $(0, 0)$, $(\pi, 0)$, $(\pi, \frac{\pi}{2})$ and $(0, \frac{\pi}{2})$.	5	CO2
Q3	Attempt any 2 parts of the followings:		
3a	1. Find the Laplace transforms of $\sin 2t \sin 3t$. 2. Find $L^{-1}(\tan^{-1} \frac{2}{s})$.	2+3=5	CO3
3b	Solve $\frac{d^2y}{dt^2} + \frac{dy}{dt} - 2y = 3 \cos 3t - 11 \sin 3t$ with $y(0) = 0$ and $\frac{dy}{dt} = 6$ at $t = 0$, using Laplace transform.	5	CO3
3c	1. Find $L[(t-1)^2 u(t-1)]$, where $u(t)$ is the unit step function. 2. Find $L^{-1}\left[\frac{1}{(s+1)(s+2)}\right]$ using convolution theorem.	2+3=5	CO3

Q4	Attempt any 2 parts of the followings:																		
4a	Find the Fourier series to represent $(x - x^2)$ in the interval $(-\pi, \pi)$. Deduce the value of $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$.	5	CO4																
4b	Find the Fourier series upto second harmonic from the following data: <table><tr><td>x</td><td>0</td><td>$\frac{\pi}{3}$</td><td>$\frac{2\pi}{3}$</td><td>π</td><td>$\frac{4\pi}{3}$</td><td>$\frac{5\pi}{3}$</td><td>2π</td></tr><tr><td>$f(x)$</td><td>1.0</td><td>1.4</td><td>1.9</td><td>1.7</td><td>1.5</td><td>1.2</td><td>1.0</td></tr></table>	x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π	$f(x)$	1.0	1.4	1.9	1.7	1.5	1.2	1.0	5	CO4
x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π												
$f(x)$	1.0	1.4	1.9	1.7	1.5	1.2	1.0												
4c	Find the half-range sine series for $f(x) = x(\pi - x)$ in the interval $(0, \pi)$ and deduce that $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \dots = \frac{\pi^2}{32}$.	5	CO4																
Q5	Attempt any 2 parts of the followings:																		
5a	Find the solution of one dimensional Heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ satisfying the boundary conditions are: (i) $u(0, t) = 0$ and $u(l, t) = 0 \forall t \geq 0$. (ii) $u(x, 0) = k \sin^3 \frac{\pi x}{l}$.	5	CO5																
5b	Solve the equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ with boundary conditions (i) $u(0, y) = 0$ and $u(\pi, y) = 0, \forall 0 \leq y < \infty$; (ii) $u(x, 0) = u_0$ and $u(x, \infty) = 0, \forall 0 < x < \pi$; where u_0 is any constant.	5	CO5																
5c	A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $y = \alpha \sin \frac{\pi x}{l}$ from which it is released at time $t = 0$. Show that the displacement of any point at a distance x from one end and at time $t > 0$ is given by $y(x, t) = a \sin \frac{\pi x}{l} \cos \frac{\pi ct}{l}$.	5	CO5																